

Figure 1.8 An object hanging from a pivot.

with the translational kinetic energy of the wheel. (c) Is the equality of parts (a) and (b) just a numerical coincidence or is it always true? (d) Would the energies be equal if the wheel were a disk rather than a hoop?

Problem 1.13 The frictional force between water and seabed in shallow seas causes an increase in the day by about 1 ms/century. Determine the torque that causes this change. Assume the Earth is a uniform sphere.

Problem 1.14 A meter stick has a pivot at one end. It is found to be in static equilibrium when acted upon by three forces that act at different points along the meter stick and act in different directions. We conclude that the net torque about the pivot is zero. Prove that the net torque about any other (arbitrary) point is also zero.

Problem 1.15 To build the pyramids it was necessary to pull heavy stones up inclined planes. Suppose a 2000 kg stone was dragged up a 20° incline at a speed of 0.25 m/s by a gang of 20 laborers. The coefficient of kinetic friction between the stone and the incline was 0.4. How much power was exerted by each laborer?

Problem 1.16 Two wooden blocks, both of mass 3 kg, are sliding in the same direction at 2 m/s on a frictionless horizontal surface. Let $M_1 = M_2 = 3$ kg and assume their speed is 2 m/s. A third block of mass 1 kg is also sliding in the same direction at a speed of 10 m/s, and it collides with the trailing 3 kg block. The third block is covered with a sticky, gooey substance, so it sticks to the trailing block. This combination block catches up with and collides *elastically* with the leading 3 kg block. Determine the final speed of the leading block. (Note that this is a one-dimensional problem. You may find it easier to solve the problem if you insert numerical values sooner rather than later.)

Problem 1.17 You are driving at 60 mph (=100 km/h). Your car's wheels have a radius of 35 cm. (a) Determine the angular velocity of the wheels. (b) What is the angular displacement of a wheel when you travel 1 km? (c) If you slow down and stop in 1 km, what is the angular acceleration of the wheel?

Problem 1.18 Aeronautical engineers have developed "tip jet" helicopters in which small jet engines are attached to the tips of the rotor. One such helicopter is powered by two ramjets. For a ramjet to develop thrust, it needs to be moving through the air quite rapidly, and is not efficient until it is moving at about 1000 km/h. Assume the rotor diameter is 10 m. Determine the angular speed of the rotor when the ramjet is moving through the air at 1000 km/h.

Problem 1.19 Figure 1.8 shows an object of mass M that is hanging from a pivot at point P. The three segments have equal length. (a) Show that the object is in equilibrium. (b) Determine whether or not this is a stable equilibrium orientation.

Problem 1.20 A solid ball of mass M and radius R rolls down an inclined plane. (a) What is its translational speed when it has descended a vertical distance h? (b) Determine its translational kinetic energy and its rotational kinetic energy.

Problem 1.21 A disk of mass 72 kg and radius 50 cm is rotating at 2000 rpm. (a) Determine its angular momentum. (b) If acted upon by a retarding force of 20 N acting tangent to the rim of the disk, determine the time required to stop the disk.

Computational Projects

The following problems can be solved easily if you know a computer language such as Matlab, Python, Fortran, or C++. A few can also be solved using a spreadsheet program such as Excel.

Although some of the computational projects in this book can be solved analytically, they will require a significant amount of brainless labor that can be done much more conveniently using a computer.

Computational Project 1.1 The position of a particle as a function of time is given by $x = 5t^3 - 2t$ (in meters). Plot the position as a function of time for the interval $t = -5$ s to $t = +10$ s. Using the relationship $\overline{v} = \Delta x/\Delta t$, obtain the average velocity at intervals of 1 second. Plot the average velocity as a function of time and, on the same graph, plot the analytical expression for the velocity.

Computational Project 1.2 The velocity of a certain particle is given by

$$v = 120\left(1 - e^{-t/10}\right) + 0.5\cos(t/2)$$

in meters per second. Plot the velocity as a function of time. Determine and plot the distance as a function of time by numerically adding up the area under the velocity vs. time curve.

Computational Project 1.3 This is a more realistic version of Problem 1.7. In that problem a teenager driving at 33.5 m/s (~75 mph) speeds past a parked policeman. The policeman and the teenager then accelerate at given constant rates and you are required to determine how far from the starting point the policeman catches up to the teenager. To make the problem somewhat more realistic, assume the teenager accelerates at $a_T = k_T e^{-b_T t}$ and the policeman accelerates at $a_P = k_P e^{-b_P t}$, where $k_T = 5.0$ mph/s and $k_P = 10$ mph/s. By plotting the positions as functions of time for various values of b_T and b_P obtain reasonable values for these constants. (Make sure your answers are reasonable. Eventually the policeman will catch the teenager, but your answer is not reasonable if the distance required is hundreds of miles!)

Computational Project 1.4 Electrons are randomly distributed in a small region of space. (The dimension is not important, but you can assume a cube with side 10 angstroms (Å).) Numerically determine the position of the center of mass, for 100, 1000, and 10 000 electrons. (You should use a random number generator to obtain the coordinates of each of the electrons.)

Problem 2.39 In cylindrical coordinates, the divergence of a vector **F** is given by

$$\nabla \cdot \boldsymbol{F} = \frac{1}{\rho}\frac{\partial(\rho F_\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}.$$

Evaluate $\nabla \cdot \boldsymbol{F}$ for

$$\mathbf{F}(\rho, \phi, z) = \rho\hat{\boldsymbol{\rho}} + z\sin\phi\hat{\boldsymbol{\phi}} + \sqrt{\rho z}\hat{\mathbf{k}}.$$

Problem 2.40 An object initially at (0,0,1) m in Cartesian coordinates, moves in the y-direction at a constant speed v. (a) Obtain an expression for its position as a function of time in spherical coordinates. (b) Determine $\dot{r}$ and $\dot{\theta}$ as functions of time. (c) Show that the speed, expressed in spherical coordinates, is constant.

Problem 2.41 A satellite moves with constant speed about the Earth in a circular orbit of radius a inclined at 90° to the equator, so that the satellite goes over the poles. A system of Cartesian coordinates fixed in the Earth has its origin at the center of the Earth, the z-axis passing through the North Pole, and the x-axis in the equatorial plane and passing through the Greenwich meridian. Obtain expressions for the position of the satellite (as functions of time) in Cartesian coordinates. The angular speed of the satellite is ω (rad/s) and the rotation rate of the Earth is Ω (rad/s).

Problem 2.42 Show that the element of arc length in spherical coordinates is

$$ds = \left(dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2\right)^{1/2}.$$

Problem 2.43 Derive the equations of transformation between cylindrical and spherical coordinates.

Problem 2.44 Show that the unit vectors of the spherical coordinate system can be expressed in terms of Cartesian coordinates as follows:

$$\hat{\mathbf{r}} = \frac{x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}}}{\left(x^2+y^2+z^2\right)^{1/2}}$$

$$\hat{\boldsymbol{\theta}} = \frac{z(x\hat{\mathbf{i}}+y\hat{\mathbf{j}})-\left(x^2+y^2\right)\hat{\mathbf{k}}}{\left(x^2+y^2\right)^{1/2}\left(x^2+y^2+z^2\right)^{1/2}}$$

$$\hat{\boldsymbol{\phi}} = \frac{-y\hat{\mathbf{i}}+x\hat{\mathbf{j}}}{\left(x^2+y^2\right)^{1/2}}.$$

Problem 2.45 An object is moving in such a way that its position in cylindrical coordinates is given by $\rho = a - bt$, $\phi = \frac{1}{2}kt^2$, $z = 0$. (a) Describe the path. (b) Find the speed as a function of time.

Problem 2.46 An object moves at constant *speed.* Prove that any nonzero acceleration must be perpendicular to the velocity.

Problem 2.47 (a) Show that the component of the acceleration in the direction of the velocity can always be expressed as

$$a_{\parallel} = \frac{\mathbf{v}\cdot\mathbf{a}}{v}.$$

(b) Find the component of the acceleration perpendicular to the velocity.

Computational Projects

Computational Project 2.1 A certain (imaginary) comet is in orbit about the Sun. It follows an elliptical path that is given by

$$r = \frac{10}{1+0.8\cos\theta},$$

where r is measured in Astronomical Units (AU). Plot the orbit of the comet in polar coordinates.

Computational Project 2.2 A cannon fires a projectile at an initial speed of $v_1 = 250$ m/s at an elevation angle of $\theta_1 = 64.1°$. After a time interval Δt, it fires another projectile at $v_2 = 300$ m/s and $\theta_2 = 47.3°$. Determine the time interval Δt such that the two projectiles collide in midair. Check your answer against the analytical result

$$\Delta t = \frac{2v_1 v_2}{g} \frac{\sin(\theta_1 - \theta_2)}{v_1 \cos\theta_1 + v_2 \cos\theta_2}.$$

Computational Project 2.3 A volcanic eruption throws a boulder vertically into the air with an initial speed of 50 m/s. (a) Write a computer program to plot the position and velocity of the boulder as a function of time. (b) Now assume the boulder is ejected at an angle of ten degrees to the vertical. Plot the two components of position and velocity as functions of time and also plot the trajectory of the boulder (on an x, y plot). Ignore the effect of air resistance.

Computational Project 2.4 In Worked Example 2.3 we obtained an expression for the launch angle that would give the greatest range for a cannon mounted on the top of a hill. (a) Validate that expression by varying the launch angle θ. (b) Determine how the optimum launch varies as the height of the hill is increased from zero.

Computational Project 2.5 Worked Example 2.6 describes a bead sliding on a helical wire. Plot the trajectory of the bead.

Computational Project 2.6 In a shotput competition an athlete throws a heavy metal ball as far as possible. A champion athlete can throw the "put" a distance of 20 m. Assume the ball is thrown with an initial speed of 13.66 m/s at an angle of 45°. If it is thrown from a height of one meter above the ground, the ball's initial position is (0,1). Using the relations

$$x = x_0 + v_{0x}t,$$
$$y = y_0 + v_{0y}t - (1/2)gt^2,$$

obtain $x(t)$ and $y(t)$. Plot the trajectory for $y > 0$. Also plot trajectories for angles ranging from 20° to 70°.

Computational Projects

Computational Project 3.1 Write a program to plot the velocity and position of the canoe of Worked Example 3.3. Assume the mass of the canoe plus person is 200 kg. Use $b = 150$ kg/s as a first estimate and get plots for various values of b. Use a reasonable choice for the initial velocity.

Computational Project 3.2 Demonstrate that the Euler–Cromer method is stable but the Euler method is not. Do this by assuming a particle moves in one dimension (x) under the influence of a force $F = -kx$. This will lead to simple harmonic motion. Compare the total energy as a function of time as obtained using the Euler method and the Euler–Cromer method. Also, on the same plot, present the total energy as a function of time as obtained from the exact (analytical) expression. You will appreciate that the Euler method does not conserve energy, the total energy obtained by the Euler–Cromer method oscillates around a constant value, whereas the analytical expression gives a constant value for the energy at all times.

Computational Project 3.3 A projectile is fired perpendicular to the Earth's surface with an initial velocity of 600 m/s. Assume a constant gravitational force acts on the projectile so its acceleration is 9.8 m/s^2 downwards. Write a program to determine the position of the particle as a function of time. Next, consider the same problem, but now include the effect of air resistance, assuming it is a retarding force that can be expressed as $f_{\text{air}} = 2.5 \times 10^{-4} v^2$, where v is the velocity of the projectile.

Problem 5.34 A particle of mass 2 kg moves along the x-axis. Its potential energy as a function of position is $V(x) = -3x + x^2$ J. (Here x is measured in meters.) The particle passes through the origin with a speed of 4 m/s. (a) Sketch the potential energy as a function of x. (b) Determine the speed of the particle when it is at $x = 1$ m. (c) Where are the turning points?

Problem 5.35 Consider the potential

$$V = -\frac{a}{r} + \frac{b}{r^2}.$$

(a) Plot the potential as a function of r. (b) Obtain an expression for the force. (c) Obtain an expression for the turning points as a function of a, b, and the total energy E.

Problem 5.36 A particle moving in a potential given by $V(x, y) = ax^2 + by^2$ is a two-dimensional oscillator with different force constants in the two directions. (This is called a nonisotropic oscillator.) Determine the motion, assuming that at time $t = 0$ the particle is passing through the origin with velocity $\mathbf{v} = v_{0x}\hat{\mathbf{i}} + v_{0y}\hat{\mathbf{j}}$.

Computational Projects

Computational Project 5.1 Assume that the potential energy is given by $V(x) = -(1/x^6) + (1/x^{12})$. Integrate Equation (5.15) numerically to determine the position of a particle of unit mass and total energy 10^{-4} energy units. Plot $x = x(t)$ and $v = v(t)$ for a particle coming in from infinity. (Infinity is approximately at $x = 5$.)

Computational Project 5.2 An asteroid of mass 5000 kg is at 40 AU from the Sun (beyond the orbit of Pluto) when it undergoes a collision with another asteroid, leaving it with zero velocity. The asteroid then slowly (at first) falls toward the Sun. Write a program to evaluate the velocity of the asteroid as it reaches the surface of the Sun. Use the Euler–Cromer algorithm. Check your result by comparing with the value obtained analytically. The purpose of this project is for you use variable time steps. Assume the radius of the Sun is 5.6×10^7 m. (1 AU = 1.5×10^{11} m).

Problem 6.30 Show that in the center of mass system, the conservation of kinetic energy from Equation (6.23) reduces to

$$\frac{P_{1\text{CM}}^2}{2\mu} = \frac{P_{1\text{CM}}'^2}{2\mu}$$

if $Q = 0$.

Problem 6.31 Show that in an elastic collision between two particles the relative velocity does not change and that it is the same in both the laboratory reference frame and the center of mass frame.

Problem 6.32 Show that the scattering angle ϕ in center of mass coordinates is related to the angle θ_2 in the laboratory coordinates by

$$\tan\theta_2 = \frac{\sin\phi}{1-\cos\phi}.$$

In other words, derive Equation (6.31).

Computational Projects

Computational Techniques: The Runge–Kutta Method

In Section 3.7 I described the Euler–Cromer algorithm for solving ordinary differential equations (ODEs). Here I will describe the Runge–Kutta method, which is more accurate and perhaps more elegant, but less transparent. It is based on the truncated Taylor series expansion for a function $g(t)$:

$$g(t+\tau) = g(t) + \tau \left.\frac{dg}{dt}\right|_{\xi}.$$

The second term is usually evaluated at t, but to make the solution more accurate, the Runge–Kutta technique evaluates it half way through the time step. That is $\xi = t + \tau/2$.

An economical way of expressing the ODE is to define two vectors, $\mathbf{x}$ and $\mathbf{f}$ in the following way. Assume a two-dimensional system so the position is given by x and y and the velocity by v_x and v_y. Then

$$\mathbf{x}(t) = [x(t) \;\; y(t) \;\; v_x(t) \;\; v_y(t)]$$
$$\mathbf{f}(\mathbf{x},t) = [v_x(t) \;\; v_y(t) \;\; a_x(t) \;\; a_y(t)]$$

and the ODE can be written

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}(t),t).$$

The second-order Runge–Kutta algorithm is obtained by first defining a new vector $\mathbf{x}^*$ by:

$$\mathbf{x}^* = \mathbf{x}^*(t+\tau/2) = \mathbf{x}(t) + \frac{1}{2}\tau\mathbf{f}(\mathbf{x}(t),t).$$

Then

$$\mathbf{x}(t+\tau) = \tau\mathbf{f}(\mathbf{x}^*, t+\tau/2).$$

It is not too difficult to appreciate that this is equivalent to the truncated Taylor series. However, the most common ODE solver is not the second-order Runge–Kutta, but the fourth-order Runge–Kutta, which is also based on the Taylor series but in a much less transparent way. In vector form, the value of $\mathbf{x}$ at time $t + \tau$ is given by

$$\mathbf{x}(t+\tau) = \mathbf{x}(t) + \frac{1}{6}\tau\left(\mathbf{F}_1 + 2\mathbf{F}_2 + 2\mathbf{F}_3 + \mathbf{F}_4\right),$$

where

$$\begin{aligned}
\mathbf{F}_1 &= \mathbf{f}(\mathbf{x}, t),\\
\mathbf{F}_2 &= \mathbf{f}\left(\mathbf{x} + \frac{1}{2}\tau\mathbf{F}_1, t + \frac{1}{2}\tau\right),\\
\mathbf{F}_3 &= \mathbf{f}\left(\mathbf{x} + \frac{1}{2}\tau\mathbf{F}_2, t + \frac{1}{2}\tau\right),\\
\mathbf{F}_4 &= \mathbf{f}\left(\mathbf{x} + \frac{1}{2}\tau\mathbf{F}_3, t + \tau\right).
\end{aligned}$$

Computational Project 6.1 A man is standing on a stationary railroad flatcar loaded with large rocks, all having the same mass of 10 kg. He is able to throw the rocks at a speed of 2 m/s, relative to himself. There are 50 rocks on the flatcar. Determine the speed of the flatcar after the man has thrown all of them straight back off the end of the car. The mass of the man plus empty flatcar is 1000 kg. Naturally, this advanced propulsion system depends on the development of completely frictionless railroad cars. Solve this problem numerically, and also solve it by using Equation (6.2). Compare your answers. Why do they not agree? Which answer do you think is correct?

Computational Project 6.2 Write a program to determine the altitude reached by a rocket launched from the surface of the Earth. Assume the rate at which fuel is burned is proportional to the amount of fuel left and the exhaust speed of the ejected gases is constant. Plot the position of the rocket as a function of time for $dM/dt = -0.01M$ if the initial mass of the fuel is 50 000 kg and $u = 2500$ m/s. The mass of the empty rocket is 10 000 kg. Ignore air resistance, but keep in mind that the gravitational force decreases with the distance from the center of the Earth.

Computational Project 6.3 A rocket is launched from the surface of the Earth. The initial mass of the rocket is 1000 kg, and it is 90 percent fuel. The exhaust gases have a speed (relative to the rocket) of 250 m/s, and the burn rate is 50 kg/s. The resistance of the air can be expressed as a retarding force given by $F = -0.5C_d\rho A v^2$, where the drag coefficient C_d is 0.35 and the density of the air can be assumed to be constant and equal to 1.20 kg/m^3. The cross-sectional area of the rocket, A, is 0.8 m^2. Do not assume $\mathbf{g}$ is constant, but you may set it equal to 9.8 m/s^2 at the ground. Determine the altitude reached by the rocket when all of its fuel is burned. Plot altitude as a function of time. Explain the shape of the curve.

Computational Project 6.4 Solve Computational Project 6.3 using a realistic profile for air density. (You can obtain tables of air density as a function of altitude using the United States Standard Atmosphere or the Smithsonian Meteorological Tables. These can be found in your library or on the Internet.)

Computational Project 6.5 A 5 MeV alpha particle is approaching a gold nucleus. The impact parameter is 1 Å. You may assume the gold nucleus is initially at rest. Plot the trajectory of the two particles and determine the angles at which both are scattered.

Computational Project 6.6 You are required to design a two-stage rocket that will accelerate a 5000 kg payload to Earth's escape velocity. Assume that 95 percent of the mass of the rockets is fuel. Assume the exhaust velocity of the rocket motors is 2000 m/s. Investigate the possible ranges of masses for the two stages and determine the configuration that will minimize the take-off weight. Determine why a single-stage rocket, burning the same amount of fuel, cannot accomplish the same objective.

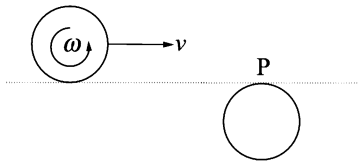


Figure 7.16 Two disks collide and stick together. See Problem 7.26.

Computational Project

Computational Project 7.1 A playground merry-go-round has been fitted with frictionless bearings. If air resistance can be ignored, this "ideal merry-go-round" will rotate indefinitely. Suppose that it is spinning at 3 rad/s when it starts to snow. The rate at which snow accumulates on the disk is 1 g/s. The disk has a radius of 1.5 m and a mass of 40 kg. Determine the angular speed of the merry-go-round after 30 min of steady snowfall. Plot ω vs. t.

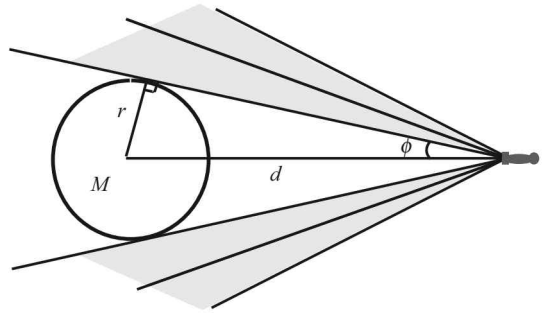


Figure 9.13 A "gravitational tractor." The shaded areas are the exhaust plumes from the rocket motors which are tilted away from the asteroid so as not to push on it. The angle ϕ is the half angle of the plume.

Problem 9.31 Show that the Minkowski metric yields $ds^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$.

Computational Projects

Computational Technique: Numerical Integration

Two simple numerical integration schemes are called the "trapezoidal rule" and "Simpson's rule." These are two applications of a more general method called the "Newton–Cotes technique." Both schemes assume you know the value of the integrand at the end points and at a number of intermediate points.

The trapezoidal rule assumes the integrand $f(x)$ varies linearly between the "data" points. Thus, if we know $f(x)$ at x_i and at x_{i+1}, then

$$\int_{x_i}^{x_{i+1}} f(x)dx = \frac{1}{2}(x_{i+1} - x_i)(f_{i+1} + f_i).$$

The total integral from initial to final values of x will be the sum of such terms. (Note that the area under the $f(x)$ curve is broken up into trapezoids. The results are often not very accurate unless the x values are very closely spaced.)

Simpson's rule uses a quadratic fit to a series of three equally spaced points, separated by distances d. Then

$$\int_{x_i}^{x_{i+1}} f(x)dx = \frac{d}{3}(f_i + 4f_{i+1} + f_{i+2}).$$

Note that the number of values of x must be odd.

Computational Project 9.1 A sphere of radius 1 m has a density that varies as $\rho(r) = 5r^3$ for $0 \leq r \leq 1$. Obtain the mass of the sphere analytically and numerically with the trapezoidal rule and Simpson's rule for nine equally spaced values of r. Compare results of the three calculations.

Computational Project 9.2 Write a computer program to determine the position and velocity of a projectile fired vertically from the Earth's surface with an initial speed of 3000 m/s. Plot the position and velocity as a function of time. The Earth is assumed to be a sphere. You may neglect air resistance, the rotation of the Earth, and the effect of any other astronomical bodies, but note that the acceleration due to gravity varies with distance from the center of the Earth.

Computational Project 9.3 Plot the potential for two equal point masses. (To make things easier, assume you are using a system of units in which the masses are unity, they are separated by unit distance and the gravitational constant is unity.) You can let the two particles lie on the x-axis.

Computational Project 9.4 Imagine a planet in the shape of a cube with side 10^5 m and density 1000 kg/m^3. Draw a Cartesian coordinate system with origin at the center of the cube and axes passing through the centers of the faces. Obtain an integral expression for the potential at a point 10^6 m from the origin along one of the axes. Write a computer program to carry out the integration and obtain a numerical answer.